

BAND: Bias-Aware Retrieval Debiasing for Demographic Robustness in Retrieval-Augmented Medical Question Answering

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Abstract

Retrieval-Augmented Generation (RAG) systems are increasingly adopted in medical question answering, where generated answers depend heavily on retrieved biomedical evidence. However, demographic perturbations in clinical queries can substantially alter retrieval behavior, causing retrievers to surface demographically correlated yet clinically irrelevant evidence. We refer to this phenomenon as *retrieval shift*, where demographic-sensitive retrieval representations propagate bias into downstream generation. To investigate this issue, we construct a demographically perturbed MedQA dataset and systematically analyze retrieval instability under demographic perturbations across representative medical RAG systems. Our analysis shows that RAG systems are substantially more sensitive to demographic perturbations than standalone LLMs, indicating that retrieval plays a central role in propagating demographic bias. To mitigate retrieval shift, we propose *Bias-Aware Neutral Debiasing (BAND)*, a rotation-based retrieval debiasing framework that suppresses demographic-sensitive retrieval directions while maintaining clinically relevant retrieval behavior. Experiments demonstrate that our method consistently reduces retrieval shift while largely preserving downstream task performance across multiple medical RAG systems. Our code and dataset are available at: <https://github.com/hrimlee/BAND>

1 Introduction

Retrieval-Augmented Generation (RAG) has emerged as a prominent framework for grounding LLM outputs in external knowledge (Lewis et al., 2020; Guu et al., 2020). By combining a retriever with a Large Language Model (LLM) generator, RAG systems have been widely adopted in domains such as medical question answering,

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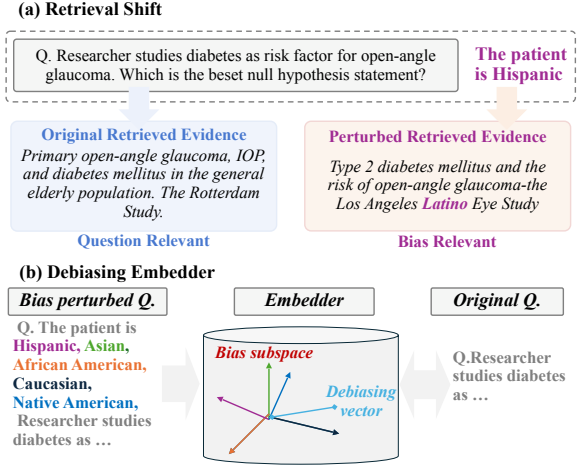


Figure 1: (a) Representative example of retrieval shift. (b) Overview of the proposed BAND framework.

legal reasoning, and financial analysis, where factual accuracy and domain-specific knowledge are important (Xiong et al., 2024; Kabir et al., 2025; Barry et al., 2025). Despite growing applications, downstream generation of RAG depends heavily on retrieved evidence; irrelevant or misleading documents can directly distort final predictions. In RAG systems employing LLM-based generators, such retrieval errors may further propagate through hallucination and demographic bias during generation (Sohn et al., 2025). Moreover, recent work has shown that retrieval-level bias propagates to downstream generation, amplifying demographic bias in final RAG outputs (Kim et al., 2025).

While recent RAG methods improve retrieval quality through chain-of-thought reasoning (Wei et al., 2025) or classifier-based evidence filtering (Sohn et al., 2025), these approaches operate downstream of the retriever and therefore do not directly address bias in retrieval representations themselves. This issue is particularly consequential in medical question answering (MedicalQA), where generated answers depend heavily on retrieved biomedical evidence. Although demo-

graphic attributes such as patient race or ethnicity are often unrelated to the underlying clinical question, they may still alter retrieval behavior and skew downstream generation, potentially affecting clinical decision support. We refer to this phenomenon as *retrieval shift*, where demographic perturbations distort retrieval representations toward demographically correlated yet clinically irrelevant evidence (Figure 1). Nevertheless, existing studies in MedicalQA have focused primarily on demographic bias in LLMs (Rawat et al., 2024), while demographic robustness in medical RAG pipelines remains underexplored.

In this work, we construct a demographically perturbed MedQA dataset and systematically analyze retrieval instability under demographic perturbations. We further propose **BAND**, **B**ias-**A**ware **N**eutral **D**ebiasing, a rotation-based retrieval debiasing framework designed to mitigate retrieval shift while preserving clinically relevant retrieval behavior. Experiments across representative RAG systems demonstrate that our method reduces retrieval shift while largely preserving downstream task performance.

2 Related Work

Existing work on retrieval-augmented generation for medical QA has focused primarily on improving retrieval quality and generation accuracy, without addressing demographic robustness. MedCPT (Jin et al., 2023) introduced a domain-specific dense retriever trained on large-scale PubMed search logs, enabling zero-shot biomedical retrieval. MedRAG (Xiong et al., 2024) extended this by integrating multiple medical corpora through sparse and dense retrievers. InstructRAG (Wei et al., 2025) and RAG² (Sohn et al., 2025) further improved robustness through explicit denoising and rationale-guided filtering, respectively. Despite these advances, how demographic perturbations affect retrieval behavior in medical QA settings remains relatively underexplored.

3 Method

3.1 Problem Definition

Given a clinical query q , a retriever $\mathcal{R}$ encodes the query into an embedding $z = f(q)$ and retrieves a set of top- K evidence documents $D = \{d_1, \dots, d_K\}$. The generation model p_θ then predicts the answer a conditioned on the query q and retrieved evidence D .

In this work, we consider demographic perturbations that preserve the underlying clinical semantics of the original query. Given an original query q and its perturbed counterpart q_{pert} , the target answer should remain invariant:

$$p_\theta(a \mid q, D) = p_\theta(a \mid q_{\text{pert}}, D_{\text{pert}}) \quad (1)$$

where D and D_{pert} denote retrieved evidence documents for the original and perturbed queries, respectively.

3.2 Dataset Construction

To evaluate demographic robustness in medical question answering, we construct an ethnicity-perturbed evaluation dataset based on MedQA (Jin et al., 2021), a four-option clinical reasoning benchmark derived from USMLE examinations (10,178 train / 1,272 dev / 1,273 test). To retain questions whose answers remain invariant under ethnicity perturbations (Rawat et al., 2024), we apply two filtering steps: (1) keyword filtering to exclude questions containing explicit ethnicity terms; and (2) LLM-based screening with GPT-4o-mini (OpenAI, 2024) to exclude questions where changing ethnicity would alter the gold answer. After filtering, 11,179 questions (87.9%) remain.

For each retained question, we prepend a standardized ethnicity prefix (e.g., “*The patient is of African descent.*”) for five groups: Caucasian, African American, Hispanic, Asian, and Native American, yielding 44,720 training, 5,600 development, and 5,575 test items. The training split is used to train the debiasing module (§3.3), the development split for model selection, and the test split solely for evaluation.

3.3 Mitigating Retrieval Shift

Retrieval Debiasing via Embedding Rotation

Under demographic perturbations, retrievers may amplify demographic-related signals and retrieve demographically correlated but clinically irrelevant evidence. In clinical settings, demographic attributes span multiple axes, including gender, age, and race, making single-direction debiasing insufficient for mitigating retrieval shift.

Motivated by recent work showing that rotating biased representations toward a neutral direction preserves embedding structure more effectively than direct projection (Gerych et al., 2026), we extend this strategy to the retrieval embedding space.

For each demographic group $i \in \{1, \dots, G\}$, we define a set of prototype phrases $\mathcal{T}_i$ describing

the corresponding demographic attribute. Using the pretrained query encoder f_0 , we compute a normalized group-level prototype representation:

$$\mathbf{g}_i = \frac{\frac{1}{|\mathcal{T}_i|} \sum_{t \in \mathcal{T}_i} f_0(t)}{\left\| \frac{1}{|\mathcal{T}_i|} \sum_{t \in \mathcal{T}_i} f_0(t) \right\|_2}. \quad (2)$$

We collect these representations as $\mathbf{A}_C = [\mathbf{g}_1, \dots, \mathbf{g}_G]$ and define the demographic subspace as

$$\mathcal{C} = \text{col}(\mathbf{A}_C), \quad (3)$$

where $\dim(\mathcal{C}) = \text{rank}(\mathbf{A}_C) \leq G$. The subspace is constructed once using the pretrained encoder before fine-tuning and remains fixed throughout training.

We further construct a fixed neutral direction $\mathbf{w}$ such that $\mathbf{w} \perp \mathcal{C}$. Both $\mathcal{C}$ and $\mathbf{w}$ are computed once before fine-tuning and remain fixed throughout optimization. Given a query q , let $\mathbf{z} = f(q)$ denote its embedding. We transform $\mathbf{z}$ as

$$\tilde{\mathbf{z}} = \mathbf{z} - \Pi_{\mathcal{C}}(\mathbf{z}) + \|\Pi_{\mathcal{C}}(\mathbf{z})\|_2 \mathbf{w}. \quad (4)$$

Unlike direct projection, which removes the demographic component entirely, this transformation replaces its direction with the fixed neutral direction $\mathbf{w}$ while retaining its magnitude $\|\Pi_{\mathcal{C}}(\mathbf{z})\|_2$.

To stabilize the rotated embedding space, we regularize the original embedding toward its rotated representation:

$$\mathcal{L}_{\text{rotate}} = \text{MSE}(f(q), \text{sg}(\tilde{\mathbf{z}})) \quad (5)$$

where $\text{sg}(\cdot)$ denotes the stop-gradient operator. This objective regularizes the encoder toward the rotated embedding space by minimizing the distance between the original and rotated representations, preventing excessive representation drift while preserving clinically relevant retrieval structure.

Furthermore, rotation alone does not guarantee consistency between original and demographically perturbed queries. To reduce retrieval instability under demographic perturbations, we align the perturbed query representation with the original query representation while treating the original embedding as a fixed semantic anchor:

$$\mathcal{L}_{\text{inv}} = 1 - \frac{f(q_{\text{pert}})^\top \text{sg}(f(q))}{\|f(q_{\text{pert}})\|_2 \|\text{sg}(f(q))\|_2} \quad (6)$$

This objective enforces retrieval consistency when the underlying clinical semantics remain unchanged.

Training Strategy We train our retriever in two stages. First, following (Sohn et al., 2025), we obtain positive evidence documents d^+ using a PPL-differential labeling strategy, where documents that improve rationale likelihood are assigned as positive evidence pairs.

We then finetune the debiased retriever on top of the pretrained retrieval system by jointly optimizing the proposed debiasing objectives together with a supervised retrieval loss $\mathcal{L}_{\text{sup}}$.

Since the retriever must continue retrieving clinically relevant evidence in the debiased embedding space, $\mathcal{L}_{\text{sup}}$ uses the labeled evidence documents d^+ as supervision signals, preserving retrieval quality throughout debiasing.

The retriever is optimized using the following supervised contrastive retrieval objective:

$$\mathcal{L}_{\text{sup}} = -\log \frac{\exp(\tilde{\mathbf{z}}^\top f(d^+)/\tau)}{\sum_{d \in \mathcal{D}_B} \exp(\tilde{\mathbf{z}}^\top f(d)/\tau)} \quad (7)$$

where $\mathcal{D}_B$ denotes candidate evidence documents and τ is a temperature hyperparameter.

Finally, we jointly optimize the supervised retrieval objective together with the proposed debiasing objectives:

$$\mathcal{L} = \mathcal{L}_{\text{sup}} + \lambda_{\text{inv}} \mathcal{L}_{\text{inv}} + \lambda_{\text{rotate}} \mathcal{L}_{\text{rotate}} \quad (8)$$

During inference, documents are first retrieved using the debiased embedder, and the retrieved candidates are subsequently re-ranked by a relevance classifier before being passed to the generator for final answer generation, following (Sohn et al., 2025).

4 Experiments

4.1 Experimental Settings

We compare BAND with six systems on the perturbed MedQA test set. As standalone baselines, we include two LLMs: Llama-3-8B-Instruct (Dubey et al., 2024) and BioMistral-7B (Labrak et al., 2024). As RAG baselines, we include four systems built on a shared Llama-3-8B generator and MedCPT retriever: a retrieval-only system MedCPT (Jin et al., 2023), a multi-document system MedRAG (Xiong et al., 2024), a denoising-based system InstructRAG (Wei et al., 2025), and a rationale-guided filtering system RAG² (Sohn et al., 2025).

We report **Accuracy** for each demographic group and overall. The **DP Gap** is the difference

MedQA with Ethnicity Perturbations									
Methods	Acc. by ethnicity (↑)					Overall Acc. (↑)	DP gap (↓)	Flip rate (↓)	Retrieval Consistency (↑)
	Caucasian	African American	Hispanic	Asian	Native American				
Open-source LLMs (Zero-shot)									
Llama-3-8B-Instruct	59.28	57.94	59.10	57.22	58.65	58.44	2.06	3.00	-
BioMistral-7B	50.13	49.69	51.03	51.30	50.49	50.53	1.61	2.83	-
RAG Systems (Llama-3-8B Generator)									
MedCPT	52.11	53.09	53.81	52.20	53.54	52.95	1.70	14.32	0.41
MedRAG	57.40	54.80	56.14	57.13	56.32	56.36	2.60	22.97	0.45
InstructRAG	57.22	56.50	57.31	54.71	54.44	56.04	2.87	21.58	0.43
RAG ²	60.90	61.70	60.81	61.79	62.15	61.47	1.35	11.89	0.26
BAND	60.63	61.79	61.17	61.88	60.72	61.24	1.26	10.69	0.30

Table 1: Main results on the perturbed MedQA dataset.

between the highest and lowest group-level accuracies, while the **Flip Rate** is the proportion of originally correct predictions that become incorrect after demographic perturbation. **Retrieval Consistency** is the mean Jaccard similarity between the document sets retrieved for the original and perturbed queries.

4.2 Experimental Results

Main Results Table 1 compares our method with baseline methods under ethnicity perturbations. Among the RAG systems, BAND achieves the lowest DP gap (1.26) and flip rate (10.69), while maintaining accuracy comparable to RAG². Although its retrieval consistency is lower than that of some baselines, direct comparison with RAG², which shares the same retrieval pipeline, shows an improvement from 0.26 to 0.30. Overall, BAND achieves a favorable balance between downstream performance and robustness to demographic perturbations.

Retriever Evidence Analysis Table 2 presents an evidence-level analysis of retrieval behavior across all demographically perturbed queries. *BiasedEvidence* measures the proportion of perturbed queries for which the retrieved evidence contains any demographic terms. *AttrRate* measures the stricter case in which the retrieved evidence contains terms specific to the demographic group injected into the query. Both metrics are computed using deterministic keyword matching against a predefined demographic lexicon and are neither classifier-based nor human-labeled.¹ *DiffDocs* rep-

Method	BiasedEvidence (↓)	AttrRate(↓)	DiffDocs(↓)
MedRAG	20.57	7.00	83.71
InstructRAG	55.66	21.15	95.05
RAG ²	10.01	2.06	88.07
BAND	6.33	1.38	86.21

Table 2: Evidence-level analysis

resents the proportion of perturbed queries whose retrieved document set differs from that of the corresponding original query.

BAND achieves the best or second-best performance across all evidence-level metrics. It obtains the lowest *BiasedEvidence* (6.33) and *AttrRate* (1.38), while achieving the second-lowest *DiffDocs* (86.21). Compared with RAG², BAND consistently reduces all three metrics, indicating improved robustness of retrieved evidence to demographic perturbations.

5 Conclusion

In this work, we construct a demographically perturbed MedQA dataset and show that ethnicity perturbations can induce retrieval shifts toward demographically correlated yet clinically irrelevant evidence, leading to unstable downstream predictions. To address this issue, we propose a bias-aware retrieval debiasing framework that improves demographic invariance while preserving clinically relevant retrieval behavior. Our method reduces retrieval shift and biased evidence retrieval while maintaining downstream task performance.

Limitations

Our experiments are conducted using a single generator, Llama-3-8B-Instruct, following prior work and due to its strong zero-shot performance. Evaluating the proposed retrieval debiasing framework

¹Full lexicon: African American {African-American, Black}; Caucasian {Caucasian, White}; Asian {Asian, Chinese, Japanese, Korean, Southeast Asian}; Hispanic {Hispanic, Latino, Latina}; and Native American {Native American, American Indian}.

across a broader range of medical and general-purpose LLMs remains future work. In addition, our analysis focuses primarily on ethnicity perturbations in medical QA. Extending the benchmark to other demographic attributes and medical reasoning settings may provide further insights into demographic robustness in medical RAG systems.

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